NAGATA COMPACTIFICATION FOR ALGEBRAIC SPACES: A SEPARATED FINITE TYPE MAP ADMITS A PROPER SURJECTIVE MODIFICATION WITH PROPER REPRESENTABLE CLOSURE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9383-9398 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 9400-9429, the complete author proof (28 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Spaces-side Nagata compactification; distinct from the Chow-type embedding card L16058 (proper blowup into projective space over a Noetherian base).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The Noetherian case of the chapter (lemma-chow-noetherian) and the Limits of Spaces approximation/descent toolkit (proposition-separated-closed-in-finite-presentation, proposition-approximate, lemmas descend-finite-presentation, descend-separated-morphism).

Source reference. Original lemma 088S: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9225–9244. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07SK: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9429–9429. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

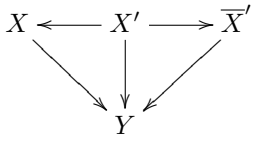
Source reference. Original lemma 0851: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9429–9429. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07SU: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9429–9429. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0873: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9429–9429. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let Y be a quasi-compact and quasi-separated algebraic space over S. Let $f : X \to Y$ be a separated morphism of finite type. Then there exists a commutative diagram*



where $X' \to X$ is proper surjective, $X' \to \overline{X}'$ is an open immersion, and $\overline{X}' \to Y$ is proper and representable morphism of algebraic spaces.

Proof. By Limits of Spaces, Proposition 0873 we can find a closed immersion $X \to X_1$ where X_1 is separated and of finite presentation over Y. Clearly, if we prove the assertion for $X_1 \to Y$, then the result follows for X. Hence we may assume that X is of finite presentation over Y.

We may and do replace the base scheme S by $\mathrm{Spec}(\mathbf{Z})$. Write $Y = \lim_i Y_i$ as a directed limit of quasi-separated algebraic spaces of finite type over $\mathrm{Spec}(\mathbf{Z})$, see Limits of Spaces, Proposition 07SU. By Limits of Spaces, Lemma 07SK we can find an index $i \in I$ and a scheme $X_i \to Y_i$ of finite presentation so that $X = Y \times_{Y_i} X_i$. By Limits of Spaces, Lemma 0851 we may assume that $X_i \to Y_i$ is separated. Clearly, if we prove the assertion for X_i over Y_i, then the assertion holds for X. The case $X_i \to Y_i$ is treated by Lemma 088S. □